REDES DE INTERNET (RI) 2024-2025

TRABALHO Nº 3 -BGP LAB: ROUTING CONFIGURATIONS

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INTRODUCTION

Welcome to the BGP for internet Configuration Lab. In this lab, you will build a topology representing an ecosystem of several Autonomous Systems, representing different Internet Service Providers (ISP). The BGP is used as exterior gateway protocol, and you will implement the routing between the AS's. You will be guided in the different steps, to create the routing policies and you will have the opportunity to have an hands-on experience taking the advantage from the BGP policies flexibility and learn some of the design best practices when applying the BGP as a protocol for the internet.

LAB OBJECTIVES AND LEARNING OUTCOMES

By the end of this lab, you will be able to:

- Follow the best practice on the implementation from BGP inside the AS and inter the AS's
- Configure BGP for internet addressing advertisement and control using the policy capabilities
- Implement BGP in an Autonomous System (AS) with a scalable approach
- Understand the basic security aspects when implementing BGP

PREREQUISITES

To successfully complete this lab, you should have:

- Basic understanding of IP routing concepts and subnetting
- Familiarity with Cisco IOS command-line interface
- Basic knowledge of BGP terminology and operations
- Experience with basic router configuration tasks

LAB ENVIRONMENT

This lab utilizes the following environment:

- Simulation Software: GNS3 (Graphical Network Simulator-3)
- Devices: Cisco routers (series 7200)
- Connections: Ethernet interfaces between routers
- Host Machines: PCs connected to specific routers for testing connectivity

Now to start GNS3 and load the lab you should:

1. Launch GNS3;
2. Go to File > Import project;
3. Select the provided .gns3 project file for this lab;
4. Wait for the project to load and verify that you see 16 routers in the topology.

EXPECTATIONS AND GUIDELINES

- Complete all tasks in each section, documenting your configurations and observations
- Answer all review questions at the end of each section
- Prepare a lab report including:
    - Your configurations for each task
    - Answers to review questions
    - Analysis of BGP behaviour observed during the lab
    - Any challenges encountered and how you resolved them
- Submit your lab report at moodle, **check the deadlines**!
- You may collaborate with classmates on understanding concepts, but all configurations and written work must be from you or your group members.
- **Note**: Plagiarism or copying configurations from other students will result in a zero grade for the lab report. All work must be your own.
- At the end of this document is a suggested template for you to use on your report.

Remember, BGP is a fundamental protocol in internet networks. The skills you develop in this lab will be directly applicable to your future work as a network engineer. Approach each task with curiosity and attention to detail, and don't hesitate to ask questions if you need clarification.

Ensure you have access to GNS3 and the required router images before beginning the lab.

Network Topology Overview

The lab has 5 main sections with 15, 15, 30, 20 e 20% weight from the total final rate. Each section includes specific tasks, verification steps, and review questions to reinforce your learning.

The main sections are:

1. Configure All the IGP configs internal to the AS's
2. iBGP and eBGP implementation without any policy or routing restrictions
3. Influence the internet routing
4. Scalability and routing simplification
5. Security practices

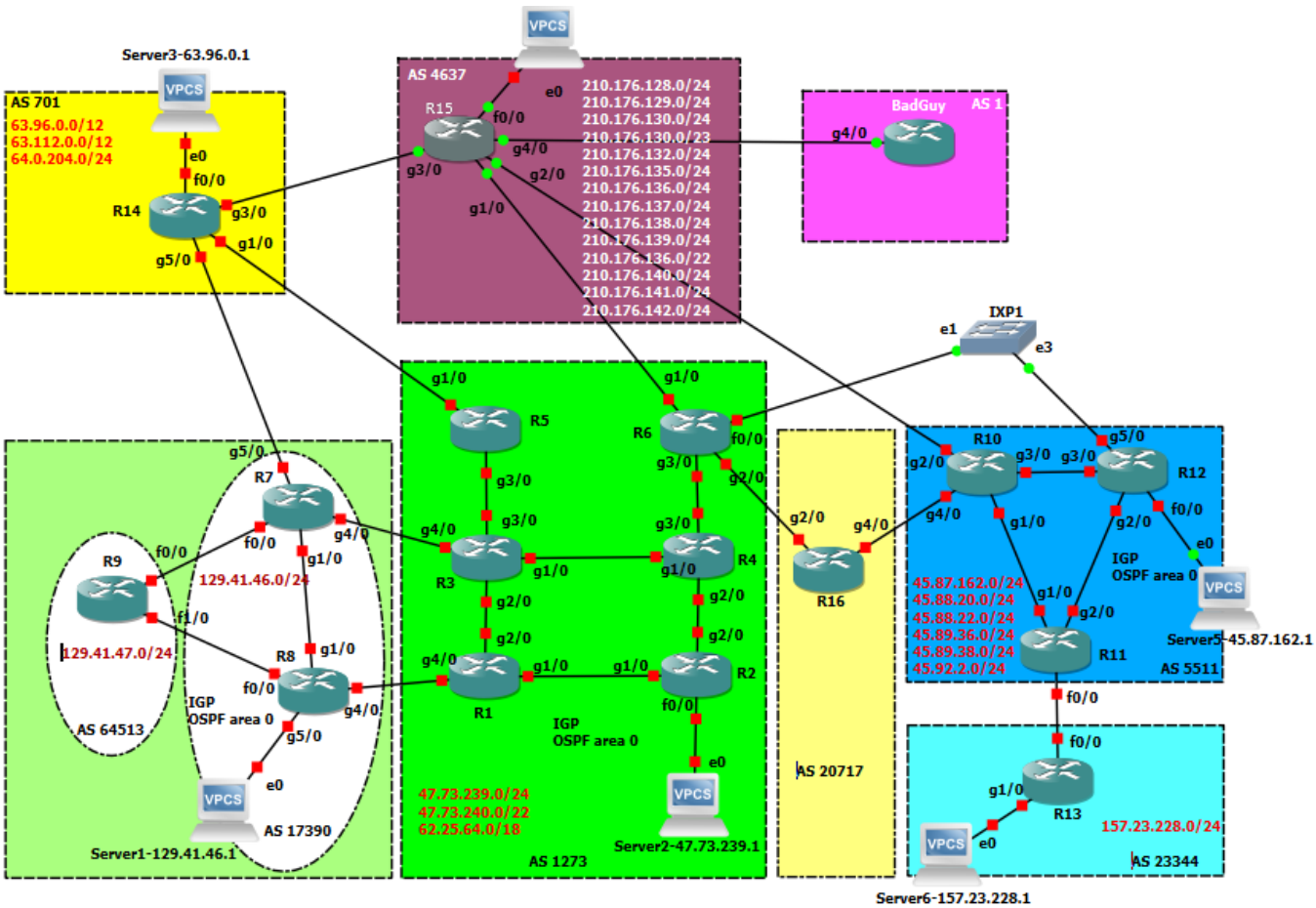


FIGURE 1 - NETWORK TOPOLOGY DIAGRAM

Lab Structure

The architecture represents a Tier 1 and Tier 2 AS's with respective downstream clients. The AS's have some public IP addresses that will be advertised inter AS's by eBGP.

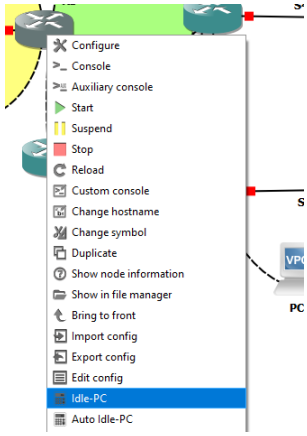
The AS 1 represents a bad deployment of BGP injecting on the internet some Bogon prefixes. With the exception from the AS1 and respective peering's, you should configure all the network topology using the engineering rules and steps.

The owner of AS 17390 decided to implement inside his network a division and attribute a private AS to R7. This part of the network is under testing, and this motivates the reason for the private AS allocation. Private ASs should not be present in the AS path attribute sent to the internet. During this lab implementation, you will need to have this in consideration and avoid propagating a private AS in your advertisements to the internet. For more info on the private AS you can consult the https://datatracker.ietf.org/doc/html/rfc6996.

Lab Rules

Generic engineering and configuration rules:

- Do not use any static routes, default routes, or policy routing unless otherwise specified.
- Save your router configuration often `#write mem`.
- Save your PC configuration often with the `VPCS> save`
- Use OSPF process id=1 (router ospf 1) in the routers where you configure OSPF.
- OSPF process should not try to establish a neighbour adjacency through interfaces where adjacencies are not expected (ex: loopbacks or connections inter AS).
- You should not send LSAs outside each OSPF domain, example AS 1273 should not send, in any situation, OSPF messages to AS 701, 4637 or 5511.
- For all the ethernet interfaces with /30 (point to point) choose the OSPF network type that will allow the fastest convergence on neighbour relations establishment.
- All the routers must have a loopback 0 (lo0) with the following convention R<ID> lo0 = 10.<ID>.<ID>.<ID> (ex: R5, loopback 0 = 10.5.5.5/32).
- All the routers must have a loopback 1 (lo1) with a public IP address from the respective AS where is part and will help on the connectivity verifications (see IP addressing from lab topology).
- Configure the router IDs with the IP address from Lo0.
- In case of high CPU on the GNS3, run the idle PC in the several routers from the topology.



- The vPCs can have in some situations some abnormal behaviour you may need to individually stop and then start the vPC.

- All the PC's and servers, represents an AS client with a public IP address from the respective AS.
- You should use the BGP next-hop-self in the iBGP sessions to resolve the next-hop from the BGP prefixes.
- iBGP sessions should use the Lo0 for the session establishment and as BGP router ID.
- On the eBGP neighbor sessions configuration, you should use the IP address from the physical interfaces between the peering routers.
- The public IP addresses should not be distributed to the IGP. Your IGP is used to advertise the private IPs from your internal AS infrastructure and allows the resolution from the BGP next hop prefixes. The iBGP sessions will distribute the public IP routes inside the AS.
- For scalability in the number of iBGP sessions, inside the AS1273, you should implement a route reflector on R3 and R4.

CONFIGURE ALL THE IGP CONFIGS INTERNAL TO THE AS'S

OBJECTIVES

In this chapter, you will:

- Configure the IP addresses for all the interfaces, as per the Table 1 - IP addressing
- Configure the OSPF as per the design rules on the AS's
- Test the private infrastructure connectivity's inside the AS's
- Test the interfaces connectivity's between the AS's
- Save this project phase as RI_24_25_GROUP<ID>_phase1

TIPS FOR THE IGP ROUTER CONFIGURATION

Each router in the topology needs basic configuration. Follow these steps for each router:

1. Right-click on the router and select "Console" to open the CLI
2. Enter privileged EXEC mode: `enable`
3. Enter global configuration mode: `configure terminal`

INTERFACE CONFIGURATION

Configure the interfaces on each router according to the IP addressing scheme provided in the topology diagram. Here's an example for R1:

1. Enter global configuration mode: `configure terminal`
2. Configure Loopback 0:

```
interface Loopback0
ip address 10.1.1.1 255.255.255.255
```

3. Configure GigabitEthernet interfaces (repeat for each interface):

```
interface GigabitEthernet0/0
ip address 10.1.2.1 255.255.255.252
no shutdown
```

4. Save the configuration:

```
end
write memory
```

Repeat this process for all routers, using the appropriate IP addresses from the topology diagram.

VERIFY BASIC CONNECTIVITY

After configuring all interfaces, verify basic connectivity:

1. Example, ping the directly connected interfaces:

```
R1# ping 10.1.2.2
```

2. Repeat similar tests between directly connected routers to ensure basic IP connectivity.
3. Verify the route path between the source and destination to confirm the routing is aligned with the requirements and your expectation

```
R1# traceroute 10.1.2.2
```

4. Inside the AS's running the IGP protocol verify the neighbour relations and the interfaces that are part of the process.

```
R1#show ip ospf neighbor

Neighbor ID     Pri   State           Dead Time   Address         Interface
10.3.3.3          0   FULL/  -        00:00:35    10.1.3.2        GigabitEthernet2/0
10.2.2.2          0   FULL/  -        00:00:30    10.1.2.2        GigabitEthernet1/0
```

```
R1#sh ip ospf interface brief
Interface    PID   Area            IP Address/Mask    Cost  State Nbrs F/C
Lo0          1     0               10.1.1.1/32        1     LOOP  0/0
Gi2/0        1     0               10.1.3.1/30        1     P2P   1/1
Gi1/0        1     0               10.1.2.1/30        1     P2P   1/1
```

5. Inside the AS's running the IGP you should be able to ping all the internal IPs including the loopback 0 from all the routers inside the AS.

6. Configure the vPC with the respective IP addresses, gateway, name and save.

```
Server1> ip 129.41.46.1/30 129.41.46.2
Server1> save
```

SAVE YOUR PROGRESS

Once you've completed the initial setup and verified connectivity:

1. Save all router configurations: `write memory` on each router and save on the vPC
2. Save your GNS3 project: File > Save project

TROUBLESHOOTING TIPS

If you encounter issues during setup:

- Verify that all interfaces are in "up/up" state using `show ip interface brief`
- Check for typos in IP addresses and subnet masks
- Ensure that you've saved configurations on all devices

- Verify the OSPF neighbours' relationship `show ip ospf summary`
- Consult the GNS3 documentation if you have software-related issues

REVIEW QUESTIONS

1. What is the concept of an Autonomous System
2. Use the https://bgp.tools/ and identify the AS's entities involved in the lab topology.
3. Identify tier 1 and tier 2 ISPs, explain the differences and the relations between them.
4. Identify in the topology, the peering's between the AS's.
5. Identify the neutral public peering interconnections in this lab topology.

BGP IMPLEMENTATION WITHOUT ANY POLICY OR ROUTING RESTRICTIONS

OBJECTIVES

In this chapter, you will:

- Establish eBGP sessions between the AS's
- Inside the AS 1273 you will establish iBGP sessions between the clients and the two route reflectors, R3 and R4 to avoid the full mesh
- The public IP addresses from the subnets used for the servers are advertised in the internet routing table
- You will have network connectivity between the ASs from any routers using the Lo1 and from any server

TIPS FOR THE BGP CONFIGURATION

1. To configure the BGP on the router you should define the respective local AS, the router-id and individually configure the neighbours

```
router bgp 701
 bgp router-id 10.14.14.14
 bgp log-neighbor-changes
 neighbor 63.112.0.2 remote-as 1273
 neighbor 63.112.0.2 soft-reconfiguration inbound
```

2. the soft-reconfiguration inbound allows you to see the individual adj-RIB-in table from each peer and will be useful to understand the prefixes that were sent by the neighbours. It will also allow you to apply the policy rules during this lab, without the need to hard reset the BGP session.

3. The BGP only advertise routes that are installed on the routing table. To allow the peer routers from an AS's to advertise the AS prefixes, you can add a static route pointing to null. You can consult the chapter Route Summarization from https://www.ciscopress.com/articles/article.asp?p=2756480&seqNum=13

```
ip route 63.96.0.0 255.240.0.0 Null0
ip route 63.112.0.0 255.240.0.0 Null0
ip route 64.0.204.0 255.255.255.0 Null0
```

and advertise the subnets via BGP with the command network. Remember that mask should exactly match the prefix from the routing table

```
router bgp 701
 network 63.96.0.114 mask 255.255.255.255
 network 63.96.0.0 mask 255.240.0.0
 network 63.112.0.0 mask 255.240.0.0
```

```
 network 64.0.204.0 mask 255.255.255.0
 no auto-summary
!
Null0
```

4. To facilitate the amount of configurations, you can define an iBGP peer-group config template and apply the peer-group config to the iBGP neighbours

```
router bgp 1273
 no synchronization
 bgp router-id 10.1.1.1
 neighbor iBGP peer-group
 neighbor iBGP remote-as 1273
 neighbor iBGP update-source Loopback0
 neighbor iBGP next-hop-self
 neighbor iBGP soft-reconfiguration inbound
 neighbor 10.3.3.3 peer-group iBGP
 neighbor 10.4.4.4 peer-group iBGP
```

5. For scalability and to avoid the full mesh in the AS1273, in the route reflectors routers, you should define the remaining neighbours as route reflectors clients

```
router bgp 1273
 no synchronization
 bgp router-id 10.3.3.3
 neighbor iBGP peer-group
 neighbor iBGP remote-as 1273
 neighbor iBGP update-source Loopback0
 neighbor iBGP route-reflector-client
 neighbor iBGP next-hop-self
 neighbor iBGP soft-reconfiguration inbound
 neighbor iBGP_RR peer-group
 neighbor iBGP_RR remote-as 1273
 neighbor iBGP_RR update-source Loopback0
 neighbor iBGP_RR soft-reconfiguration inbound
 neighbor 10.1.1.1 peer-group iBGP
 neighbor 10.2.2.2 peer-group iBGP
 neighbor 10.4.4.4 peer-group iBGP_RR
 neighbor 10.5.5.5 peer-group iBGP
 neighbor 10.6.6.6 peer-group iBGP
```

VERIFICATIONS TIPS AND CHECKS

The commands and checks are indicative, you should run in the different routers from the topology and interpret the output to confirm if you are getting the expected result (ex: prefixes that you expect, number of peering's and respective status).

1. Verify the BGP peer status

```
R1#sh ip bgp summ
BGP router identifier 10.1.1.1, local AS number 1273
BGP table version is 100, main routing table version 100
55 network entries using 7260 bytes of memory
111 path entries using 5772 bytes of memory
12/10 BGP path/bestpath attribute entries using 2016 bytes of memory
8 BGP rrinfo entries using 192 bytes of memory
8 BGP AS-PATH entries using 192 bytes of memory
0 BGP route-map cache entries using 0 bytes of memory
0 BGP filter-list cache entries using 0 bytes of memory
Bitfield cache entries: current 3 (at peak 4) using 96 bytes of memory
BGP using 15528 total bytes of memory
BGP activity 55/0 prefixes, 615/504 paths, scan interval 60 secs

Neighbor        V           AS MsgRcvd MsgSent   TblVer  InQ OutQ Up/Down  State/PfxRcd
10.3.3.3        4         1273     260     242      100    0    0 03:34:47        49
10.4.4.4        4         1273     242     224      100    0    0 03:34:35        49
47.73.240.2     4        17390     565     335      100    0    0 05:17:31         8
```

2. Verify the prefixes received and sent from or to a determined neighbour

```
R3#show ip bgp neighbors 10.6.6.6 received-routes
BGP table version is 39, local router ID is 10.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*>i45.87.162.0/24   10.6.6.6                 0    100      0 20717 5511 i
*>i45.88.20.0/24    10.6.6.6                 0    100      0 20717 5511 i
*>i45.88.22.0/24    10.6.6.6                 0    100      0 20717 5511 i
*>i45.89.36.0/24    10.6.6.6                 0    100      0 20717 5511 i
*>i45.89.38.0/24    10.6.6.6                 0    100      0 20717 5511 i
*>i45.92.2.0/24     10.6.6.6                 0    100      0 20717 5511 i
(...)


R3#show ip bgp neighbors 10.6.6.6 advertised-routes
BGP table version is 117, local router ID is 10.3.3.3
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*>i10.0.0.0         10.6.6.6                 0    100      0 4637 1 ?
*>i45.87.162.0/30   10.6.6.6                 0    100      0 5511 i
*>i45.87.162.0/24   10.6.6.6                 0    100      0 5511 i
```

```
*>i45.87.162.110/32 10.6.6.6                0    100      0 5511 i
*>i45.92.2.0/24     10.6.6.6
```

3. Verify the prefixes received and sent from or to a determined neighbour

Verify the prefixes installed on the Loc-RIB BGP table

```
R9#show ip bgp
BGP table version is 46, local router ID is 10.9.9.9
Status codes: s suppressed, d damped, h history, * valid, > best, i - internal,
              r RIB-failure, S Stale
Origin codes: i - IGP, e - EGP, ? - incomplete

   Network          Next Hop            Metric LocPrf Weight Path
*  45.87.162.0/24   129.41.46.5                            0 17390 1273 20717 5511 i
*>                  129.41.46.9                            0 17390 1273 20717 5511 i
*  45.88.20.0/24    129.41.46.5                            0 17390 1273 20717 5511 i
*>                  129.41.46.9                            0 17390 1273 20717 5511 i
*  45.88.22.0/24    129.41.46.5                            0 17390 1273 20717 5511 i
*>                  129.41.46.9                            0 17390 1273 20717 5511 i
*  45.89.36.0/24    129.41.46.5                            0 17390 1273 20717 5511 i
*>                  129.41.46.9                            0 17390 1273 20717 5511 i
*  45.89.38.0/24    129.41.46.5                            0 17390 1273 20717 5511 i
(…)
```

4. On the end of this chapter, you should be able to run the TCLSH (Tool Control Language) Shell for testing. When the BGP will be in place, from all the routers, using the Lo1 public IP addresses as source address, you should be able to ping all the Lo1 interfaces from the other routers and the public IPs from the servers. See the TCLSH script on the appendix TCLSH (Tool Control Language) Shell for testing.

SAVE YOUR PROGRESS

1. Save all router configurations: `write memory` on each router
2. Save this project phase as RI_24_25_GROUP<ID>_phase2

REVIEW QUESTIONS

1. You applied the configuration next-hop-self on the iBGP peering explain why
2. How are the route prefixes propagated inside the AS when you have the route reflectors configured?
3. How is the BGP next hop reachability solved inside an AS?
4. What is the behaviour from BGP when the prefix next-hop is not reachable
5. Why is a good practice the loopback IP address utilization in the iBGP sessions?
6. Observe in the Wireshark and add to the report a trace from a BGP session establishment and described the packets and respective relation with the following stats.

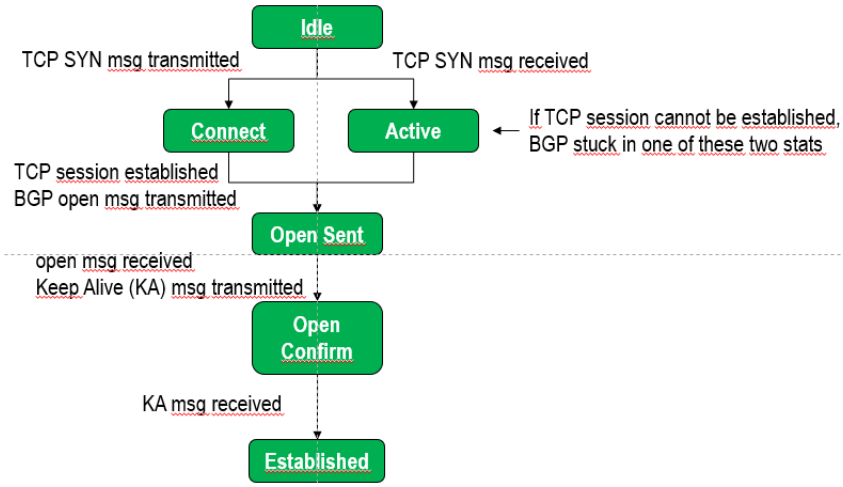


FIGURE 2 – BGP STATES

SCALABILITY AND ROUTING SIMPLIFICATION

OBJECTIVES

In this chapter, you will:

- Whenever possible you should aggregate the IP prefixes and advertise only the aggregate route on the eBGP sessions. We should not have advertised prefixes longer than /24 in the internet routing table
- The internet peering's should accept a maximum of 50 prefixes
- We should not have private ASs in the middle from an AS path attribute
- The AS 23344 has only one peering to the internet and want to receive only the default route from the eBGP peering

TIPS FOR THE BGP CONFIGURATION

1. You should verify in each AS, the possibility to summarize the prefixes. When possible, the command aggregate will allow to advertise the summarized prefix only. You can apply this approach on the AS's with one peer router only

```
router bgp 64513
 aggregate-address 129.41.47.0 255.255.255.0 summary-only
```

2. On the AS's with more than one router, the Lo1 are advertised in the iBGP and you need these prefixes for the internal routing. You can use a prefix-list to advertise to the eBGP peers only the prefixes with mask /24 or lower.

```
ip prefix-list PREF_LIST_ADV_SUMM seq 5 permit <prefix add/mask> le <24>
…
router bgp 1273
  neighbor 47.73.240.6 prefix-list PREF_LIST_ADV_SUMM out
```

3. After changing or applying an out peering policy (ex: prefix-list application, route-map), you need to soft restart the peer to apply the policy changes

```
R3#clear ip bgp 47.73.240.6 soft out
```

4. You can check the AS path attribute from the prefixes using the following commands

```
R3#show ip bgp 210.176.129.0
R3#show ip bgp
```

This will help you to see the prefixes with private AS's, you can remove the private AS's from the AS path applying the following command on the correct routers and peering's

```
neighbor 129.41.46.10 remove-private-as
```

SAVE YOUR PROGRESS

REVIEW QUESTIONS

1. When you applied a prefix-list or route-map in a peering neighbour what is the table that reflects the information sent on the updates
2. Refer the AS path private range and some use cases where can be useful to use this range
3. Why is the control from the number of prefixes advertised to the internet a good practice

a) Assuming you succeed in this configuration explain, based on the BGP prefix selection algorithm rules, the selection for the prefix 64.0.204.0 in R12. Use the following explanation to understand the selection. (http://www.cisco.com/en/US/tech/tk365/technologies_tech_note09186a0080094431.shtml)

```
R12#sh ip bgp 64.0.204.0
BGP routing table entry for 64.0.204.0/24, version 18717
Paths: (2 available, best #2, table Default-IP-Routing-Table)
  Advertised to update-groups:
        2
  4637 701, (received & used)
    10.10.10.10 (metric 2) from 10.10.10.10 (10.10.10.10)
      Origin IGP, metric 0, localpref 100, valid, internal
  1273 701, (received & used)
    47.73.240.17 from 47.73.240.17 (10.6.6.6)
      Origin IGP, localpref 100, valid, external, best
```

INFLUENCE THE INTERNET ROUTING

OBJECTIVES´

In this chapter, you will:

- Using the BGP policies and attributes, the AS20717 should be preferred for the originated traffic from AS 1273 and AS AS511, you should assure this rule in both directions.
  Example:
  - Between AS17390 <> AS1273 <> IXP1 <> AS5511 <> AS23344
  - Between AS1273 <> AS20717 <> AS5511

In case you need to use regular expressions, the following link can help to test the logics https://regex101.com/ . The PCH - BGP Best Practices for ISPs.pdf has some examples from most common expressions.

SAVE YOUR PROGRESS

REVIEW QUESTIONS

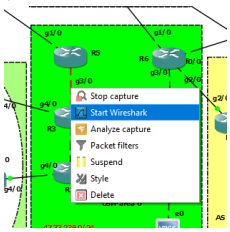
1. Describe in your report the policy options you used to implement the routing logics. On the report you should clearly describe the objectives, configs (ex: prefix-list, route-map etc) and the report will be your support on the final discussion.

SECURITY PRACTICES

OBJECTIVES

- Activate the authentication on all the eBGP peering's on the AS701. This a good practice for all the peers nevertheless for the lab proposes will be ok to test on this AS only.
- The BadGuy router is advertising Bogons. Filter the Bogon prefixes on the peer routers from AS 1273 (see "PCH - BGP Best Practices for ISPs.pdf" from CD alunos 2).
- Remote Triggered Black Hole (RTBH) filtering is a popular and effective technique for the mitigation of denial-of-service (DoS) attacks. The prefix 63.96.0.114 was identified as a bot (https://www.cloudflare.com/en-gb/learning/bots/what-is-a-bot/) and doing a DoS to several destinations on the AS1273. Use the R4 as trigger router to filter the prefix 63.96.0.114 in this AS and avoid the attack. For testing and validation from your implementation you can configure on the R14 a Lo2 with the IP 63.96.0.114.
  See the https://www.cisco.com/c/dam/en_us/about/security/intelligence/blackhole.pdf for a complete understanding from this technic.

```
interface Loopback2
 ip address 63.96.0.114 255.255.255.255
```

  - Ping from R14 with source interface Lo2 from R14 to Server2
  - Verify the traffic through R5-R3, you can start a Wireshark trace on the interface and filter the ICMP packets
  - 

  - R4 will be the trigger router that will propagate, inside the AS, the Bot detected prefix.

```
R4#
ip route 63.96.0.115 255.255.255.255 Null0 tag 66
router bgp 1273
 redistribute static
 neighbor iBGP send-community
 neighbor iBGP route-map black-hole-trigger out
...
ip prefix-list BH seq 5 permit 63.96.0.115/32
...
route-map black-hole-trigger permit 10
 match ip address prefix-list BH
 set local-preference 200
 set origin igp
 set community no-export
 set ip next-hop 192.0.2.1
```

```
!
route-map black-hole-trigger permit 20
```

- After applying this in the R4 (trigger router) all the edge routers from the AS 1273 must have a static route for an unused IP address space. For example, 192.0.2.1/32 is set to Null0. The IP address 192.0.2.1 is reserved for use in test networks and is not used as a deployed and apply the uRPF (unicast Reverse Path Forwarding) in the peering interfaces.

```
ip route 192.0.2.1 255.255.255.255 Null0
...
interface GigabitEthernet1/0
 ip verify unicast reverse-path
```

SAVE YOUR PROGRESS

REVIEW QUESTIONS

1. Describe in your report the implementation from the RTBH inside the AS1273, as suggestion build a drawing similar with the explanation from the ones inside the https://www.cisco.com/c/dam/en_us/about/security/intelligence/blackhole.pdf to explain how you did the implementation in your lab.
2. Explain the uRPF behaviour and the role filtering the traffic source address.

CONCLUSION

Congratulations on completing the BGP Configuration Lab! Throughout this lab, you've gained hands-on experience with:

- Configuring BGP networks
- Understand the concepts from internet topology and routing
- Secure BGP AS's
- Optimizing BGP networks and use some of the basic attributes to influence the routing
- Troubleshooting common BGP issues

These skills form the foundation of designing, implementing, and maintaining BGP networks in real-world scenarios.

APPENDICES

IP ADDRESSING FROM LAB TOPOLOGY

TABLE 1 - IP ADDRESSING

	Interface	Private IP add	Public IP add
R1	Lo0	10.1.1.1/32	
	Lo1		47.73.239.11/32
	g1/0	10.1.2.1/30	
	g2/0	10.1.3.1/30	
	g4/0		47.73.240.1/30
R2	Lo0	10.2.2.2/32	
	Lo1		47.73.239.22/32
	f0/0		47.73.239.2/30
	g1/0	10.1.2.2/30	
	g2/0	10.2.4.1/30	
R3	Lo0	10.3.3.3/32	
	Lo1		47.73.239.33/32
	g1/0	10.3.4.1/30	
	g2/0	10.1.3.2/30	
	g3/0	10.3.5.1/30	
	g4/0		47.73.240.5/30
R4	Lo0	10.4.4.4/32	
	Lo1		47.73.239.44/32
	g1/0	10.3.4.2/30	
	g2/0	10.2.4.2/30	
	g3/0	10.4.6.1/30	
R5	Lo0	10.5.5.5/32	
	Lo1		47.73.239.55/32
	g1/0		63.112.0.2/30
	g3/0	10.3.5.2/30	
R6	Lo0	10.6.6.6/32	
	Lo1		47.73.239.66/32
	f0/0		47.73.240.17/30
	g1/0		47.73.240.13/30
	g2/0		47.73.240.21/30
	g3/0	10.4.6.2/30	
R7	Lo0	10.7.7.7/32	
	Lo1		129.41.46.77/32
	f0/0		129.41.46.9/30
	g1/0	10.7.8.1/30	
	g4/0		47.73.240.6/30

	g5/0		63.112.0.6/30
R8	Lo0	10.8.8.8/32	
	Lo1		129.41.46.88/32
	f0/0		129.41.46.5/30
	g1/0	10.7.8.2/30	
	g4/0		47.73.240.2/30
	g5/0		129.41.46.2/30
R9	Lo0	10.9.9.9/32	
	Lo1		129.41.47.99/32
	f0/0		129.41.46.10/30
	f1/0		129.41.46.6/30
R10	Lo0	10.10.10.10/32	
	Lo1		45.87.162.110/32
	g1/0	10.10.11.1/30	
	g2/0		210.176.129.2/30
	g3/0	10.10.12.1/30	
	g4/0		45.88.20.1/30
R11	Lo0	10.11.11.11/32	
	Lo1		45.87.162.111/32
	f0/0		45.88.20.5/30
	g1/0	10.10.11.2/30	
	g2/0	10.11.12.1/30	
R12	Lo0	10.12.12.12/32	
	Lo1		45.87.162.112/32
	f0/0		45.87.162.2/30
	g2/0	10.11.12.2/30	
	g3/0	10.10.12.2/30	
	g5/0		47.73.240.18/30
R13	Lo0	10.13.13.13/32	
	Lo1		157.23.228.113/32
	f0/0		45.88.20.6/30
	g1/0		157.23.228.2/30
R14	Lo0	10.14.14.14/32	
	Lo1		63.96.0.114/32
	f0/0		63.96.0.2/30
	g1/0		63.112.0.1/30
	g3/0		63.112.0.9/30
	g5/0		63.112.0.5/30
R15	Lo0	10.15.15.15/32	
	Lo1		210.176.128.115/32
	f0/0		210.176.128.2/30
	g1/0		47.73.240.14/30
	g2/0		210.176.129.1/30

	g3/0		63.112.0.10/30
	g4/0		210.176.129.5/30
R16	Lo0	10.16.16.16/32	
	Lo1		
	g2/0		47.73.240.22/30
	g4/0		45.88.20.2/30
Badguy	Lo0	10.66.66.66/32	
	g4/0		210.176.129.6/30
Server1	e0		129.41.46.1/30
Server2	e0		47.73.239.1/30
Server3	e0		63.96.0.1/30
Server4	e0		210.176.128.1/30
Server5	e0		45.87.162.1/30
Server6	e0		157.23.228.1/30

TCLSH (TOOL CONTROL LANGUAGE) SHELL FOR TESTING.

```
R9#tclsh
R9(tcl)#
foreach address {
47.73.239.11
47.73.239.22
47.73.239.33
47.73.239.44
47.73.239.55
47.73.239.66
129.41.46.77
129.41.46.88
129.41.47.99
45.87.162.110
45.87.162.111
45.87.162.112
157.23.228.113
63.96.0.114
210.176.128.115
129.41.46.1
47.73.239.1
63.96.0.1
210.176.128.1
45.87.162.1
157.23.228.1
} {ping $address source lo1}
```

DOCUMENTATION

https://bgp4.as/looking-glasses/

https://bgp.tools/

https://www.ciscopress.com/articles/article.asp?p=2756480

https://www.cisco.com/c/dam/en_us/about/security/intelligence/blackhole.pdf

https://www.cisco.com/c/en/us/support/docs/ip/border-gateway-protocol-bgp/200153-BGP-Route-Reflection-and-Multiple-Cluste.html

BGP Tutorial.pdf

PCH - BGP Best Practices for ISPs.pdf